QP CODE: 25900297

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION October 2025

THIRD SEMESTER

MG3DSCCND202-DIET AND HEALTH

2024 ADMISSION ONWARDS

Duration: 2 Hours Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple ChoiceQuestions

Answer all questions

Each question carries 1 mark

1. b - Assessing obesity

2. b - Not eating enough because of fear of gaining weight

3. b - Heart attack

4. c - smoking

5. a - Fibrosis and inflammation

6. b - low protein low sodium diet

Part B

Short Answer

Answer any 2 questions.

Each question carries 2 marks.

7. List two lifestyle modification strategies used in the prevention of

Obesity

maintaining a healthy diet - **1 mark**

increasing physical activity - **1 mark**

8. List any two advantages of DASH diet

1. Reduces blood pressure - **1 mark**

2. promotes heart health - **1 mark**

9.Write two common symptoms of hepatitis.

1. fatigue - **1 mark**

2. jaundice / nausea/ diarrhea / fever - **1 mark**

10. List any two common causes of kidney stones.

1. dehydration / intake of oxalate or calcium rich / purine rich food / high sodium intake/ certain medications / family history - **2 marks**

Part C

Short Essay

Answer any 4 questions

Each question carries 5 marks.

11. What are the role of lifestyle modification as supportive therapy in

diabetes management?

1. Improve blood glucose control, Weight management, Reduce risk of complications, Blood pressure and cholesterol control, Reduce cardiovascular risk / manage stress and promote well being **- 5 marks**

12. Explain the classification of hypertension.

Normal - 120/80 mmHg

Elevated - 120 - 129 / 80 mmHg

Stage 1 - 130 - 139 /80-89 mmHg

Stage 2 - 140/90 mmHg

Hypertensive crisis - 180/120 mmHg - **5 marks**

13. Describe the pathogenesis and diagnostic criteria of rheumatic heart

Disease.

- Causative agent group A Beta - hemolytic streptococci
- Untreated strep throat
- Rheumatic fever

- All layers of the heart and mitral valve become inflamed
- Vegetation forms
- Valvular regurgitation and stenosis
- Heart failure - **3 mark**

Diagnostic criteria - fever / throat pain/ chest pain/ carditis

Diagnosis - ECG / MRI / USS/ X RAY - **2 marks**

14. Compare the advantages and disadvantages of hemodialysis.

Advantages -

effective waste removal , rapid correction, Medical supervission, Scheduled treatment / treatment frequency, Access to medical resources - **2.5 marks**

Disadvantages -

Time consuming , Travel required, Risk of infection, Strict dietary and fluid restrictions, Potential side effects, Expensive - **2.5 marks**

15.List the viral and chemical agents that cause liver damage.

Industral chemicals, Environmental toxins, Natural toxins, Alcohol, Sugar salt, Drugs and chemicals - **5 marks**

16. Explain the causes and types of acute renal failure

Causes -

- Interruption of blood flow to the kidneys from severe injury or illness
- Kidney inflammation due to toxins, drugs, infections or reduced blood supply

- Obstruction in urinary tract
- Kidney stones
- Nephrosclerosis - **2.5 marks**

Types -

- Prerenal
- Intra renal
- Post renal /

- Initiation phase, oliguric phase, diuretic phase, recovery phase - **2.5 marks**

Part D

Essay

Answer any 2 questions

Each question carries 10 marks

17. Explain the pharmacological and non-pharmacological treatment of diabetes mellitus

Introduction - **3 marks**

Non - Pharmacological treatement -

Dietary management

Physical activity, Weight management, Blood sugar monitoring, Self monitoring, Stress and, sleep management **- 4 marks**

Pharmacological treatement -

Insuline therapy

Biguanides - eg metformin - increase insuline

Sulphonylureas - stimulate pancreatic insuline secreation

Meglitinides - stimulate rapid insuline secreation

Thiazolidinediones - improve insulin sensitivity in muscle and adipose tissue **- 3 marks**

18. Explain the types and aetiology of diabetes mellitus

Type 1 - juvenile onset diabetes or insulin dependent dm - autoimmune disease, acidosis is common, destruction of insulin producing B - cells of pancreas,underweight.

Type 2 - non insulin dependent, adult onset insulin production is impaired acidosis is infrequent, usually overweight, insulin resistance - **5 marks**

Aetiology - genetics , environmental factors, immunological factors, high carbohydrates intake, high caffeine intake, low fiber diet, overweight, sedentary lifestyle, etc. - **5 marks**

19. Discuss the management and preventive strategies for hypercholesterolemia in detail.

Introduction - **3 marks**

Healthy diet, reduce saturated and unsaturated fats, increase soluble fiber intake, limit sodium and added sugar , regular physical activity, maintain a healthy weight, avoid smoking and alcohol usage, manage stress. Regular monitoring , lifestyle modification - **7 marks**

20. Outline in detail the causes, symptoms, complications, and dietary management of chronic renal failure.

Introduction - **2 marks**

Causes - glomerulonephritis, family hystory, nephrosclerosis, kidney stones, diabetes mellitus, contineuos use of medications hypertension, unhealthy dietary habits, dehydration, sedentory lifestyle, liver disease etc - **2 marks**

Symptoms- fatigue, weakness, oliguria, weightloss, bone pain, sleeping disturbance, menstrual periods stops , abdominal pain etc. -**2 marks**

Complications - edema, heart disease, increased risk of bone fractures, anemia, reduced fertility, seizures, memory lose, personality changes, weakend immunity, pregnency complications, electrolyte and acid - base imbalances, endocrine and metabalic complications, gastrointestinal complications etc. - **2 marks**

Dietary management - adequate calorie, restricted protein, restricted sodium, restricted potasium, restricted phosphorus, adeqate calsium, fluid restriction, use healthy fats, normal carbohydrates, avoid excessive vitamin A - **2 marks**